

第三章质量评估

1. A 2. A 3. B 4. D

5. A [解析] $\because$ 将 $\triangle OAB$ 绕点 O 逆时针旋转 80° , 得到 $\triangle OCD$,

$$\therefore \angle AOC = \angle BOD = 80^\circ.$$

$$\therefore \angle BOC = 110^\circ,$$

$$\therefore \angle AOB = \angle BOC - \angle AOC = 110^\circ - 80^\circ = 30^\circ.$$

$$\therefore \angle AOD = \angle BOD - \angle AOB = 80^\circ - 30^\circ = 50^\circ.$$

$$\therefore \angle \alpha = 50^\circ.$$

故选 A.

6. C [解析] $\because$ 将 $\triangle ABC$ 沿着射线 BC 向右平移得到 $\triangle DEF$,

$$\therefore AD = BE = CF.$$

$$\therefore AD = 2CE, CF = 2,$$

$$\therefore BE = 2, CE = 1,$$

$$\therefore BC = BE + CE = 3. \text{ 故选 C.}$$

7. C

8. A [解析] 连接 PP' .

由旋转可得 $\angle PAP' = \angle CAB = 60^\circ, AP = AP'$,

$\therefore \triangle APP'$ 是等边三角形,

$$\therefore PP' = AP = 4,$$

$$\therefore S_{\triangle APP'} = \frac{1}{2} \times 4 \times 2\sqrt{3} = 4\sqrt{3}.$$

由旋转可得 $BP' = CP = 5$,

$$\therefore BP^2 + P'P^2 = 25 = P'B^2,$$

$\therefore \triangle BPP'$ 是直角三角形, 且 $\angle BPP' = 90^\circ$,

$$\therefore S_{\triangle BPP'} = \frac{1}{2} \times 3 \times 4 = 6,$$

$\therefore$ 四边形 $AP'BP$ 的面积为 $4\sqrt{3} + 6$.

故选 A.

9. (2, -3)

10. (-6, -2) [解析] $\because$ 点 $P(m, m+4)$ 向上平移 2 个单位长度到点 Q ,

$$\therefore Q(m, m+6).$$

$\because$ 点 Q 在 x 轴上,

$$\therefore m+6=0, \text{ 解得 } m=-6,$$

$$\therefore \text{点 } P(-6, -2).$$

故答案为 (-6, -2).

11. 55

12. 5 [解析] 由题图知, $\triangle A'B'C'$ 是由 $\triangle ABC$ 先向右平移 3 个单位长度, 再向上平移 2 个单位长度得到的,

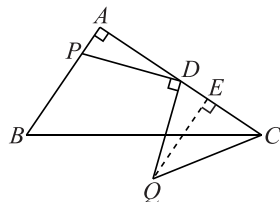
$$\therefore m=3, n=2,$$

$$\therefore m+n=5.$$

故答案为 5.

13. $\frac{39}{2}$

14. $3\sqrt{2}$ [解析] 如图所示, 过点 Q 作 $QE \perp AC$ 于点 E , 则 $\angle A = \angle QED = 90^\circ$.



由旋转可得, $DP = QD, \angle PDQ = 90^\circ$,

$$\therefore \angle ADP = \angle EQD.$$

在 $\triangle DAP$ 和 $\triangle QED$ 中,

$$\therefore \angle ADP = \angle EQD, \angle A = \angle QED, DP =$$

QD ,

$$\therefore \triangle DAP \cong \triangle QED (\text{AAS}),$$

$$\therefore QE = AD = \frac{1}{2}AC = 3, DE = AP,$$

$\therefore$ 点 Q 在平行于 AC 的直线上运动.

当点 E 与点 C 重合时, CQ 的值最小, CQ 的最小值为 3,

此时 $AP = DE = 3$,

$$\therefore DP = \sqrt{AP^2 + AD^2} = \sqrt{3^2 + 3^2} = 3\sqrt{2}.$$

故答案为 $3\sqrt{2}$.

15. 解: (1) 由平移可知 $\angle ACB = \angle F = 26^\circ$,

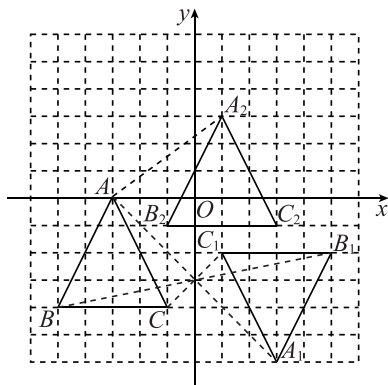
$$\therefore \angle A = 180^\circ - \angle B - \angle ACB = 180^\circ - 74^\circ - 26^\circ = 80^\circ.$$

(2) $\because BC = 3 \text{ cm}, EC = 2 \text{ cm},$

$$\therefore BE = BC - EC = 3 - 2 = 1 (\text{cm}),$$

即 $\triangle ABC$ 平移的距离为 1 cm.

16. 解: (1) 如图, $\triangle A_1B_1C_1$ 为所求作.



(2) 如图, $\triangle A_2B_2C_2$ 为所求作.

如图, 连接 AA_2 , 由图可知 AA_2 的长即为平移的距离, 由勾股定理, 得 $AA_2 = \sqrt{3^2 + 4^2} = 5$, 即平移的距离是 5 个单位长度.

17. 解: (1) $\because BD \perp AC$,

$$\therefore \angle BDC = 90^\circ.$$

$\because$ 将线段 CD 绕点 C 顺时针旋转 α 后得到线段 CE ,

$$\therefore CD = CE, \angle DCE = \alpha,$$

$$\therefore \angle DCE = \angle ACB.$$

$$\because CA = CB,$$

$$\therefore \triangle ACE \cong \triangle BCD (\text{SAS}),$$

$$\therefore \angle E = \angle BDC = 90^\circ.$$

(2) 由(1)知 $CD = CE = 2$,

$$\therefore AC = AD + CD = 3 + 2 = 5.$$

$$\because CA = CB,$$

$$\therefore CB = 5.$$

$$\text{在 Rt } \triangle BCD \text{ 中, } BD = \sqrt{BC^2 - CD^2} = \sqrt{5^2 - 2^2} = \sqrt{21}.$$

18. 解: (1) $\because \triangle ABC$ 是等边三角形,

$$\therefore \angle ACB = 60^\circ.$$

$$\because AD = CD, \angle ADC = 120^\circ,$$

$$\therefore \angle DAC = \angle DCA = (180^\circ - 120^\circ) \div 2 = 30^\circ,$$

$$\therefore \angle BCD = \angle ACD + \angle ACB = 30^\circ + 60^\circ = 90^\circ.$$

(2) $\because \triangle ABD$ 绕点 A 逆时针旋转 60° 得到 $\triangle ACD'$, $\angle ADB = 150^\circ, BD = 8$,

$$\therefore \angle AD'C = \angle ADB = 150^\circ, AD = AD',$$

$$\angle DAD' = 60^\circ, BD = CD' = 8,$$

$\therefore \triangle ADD'$ 是等边三角形,

$$\therefore DD' = AD = 4, \angle AD'D = 60^\circ,$$

$$\therefore \angle DD'C = \angle AD'C - \angle AD'D = 150^\circ - 60^\circ = 90^\circ,$$

$$\therefore CD = \sqrt{DD'^2 + CD'^2} = \sqrt{4^2 + 8^2} = 4\sqrt{5},$$

$\therefore CD$ 的长为 $4\sqrt{5}$.

19. 解: (1) $\because$ 将 $\triangle ACD$ 绕点 A 按顺时针方向旋转 60° 得到 $\triangle ABE$,

$$\therefore \angle DAE = 60^\circ, \angle ADC = \angle AEB.$$

$$\because \angle AFE = \angle BFD = 97^\circ,$$

$$\therefore \angle AEB = 180^\circ - \angle DAE - \angle AFE = 23^\circ,$$

$$\therefore \angle ADC = 23^\circ,$$

即 $\angle ADC$ 的度数是 23° .

(2) 如图, 连接 DE .

$$\because \angle BDC = 7^\circ, \angle ADC = 23^\circ,$$

$$\therefore \angle ADB = \angle BDC + \angle ADC = 30^\circ.$$

由旋转得 $AD = AE, \angle DAE = 60^\circ$,

$\therefore \triangle ADE$ 是等边三角形,

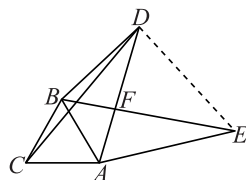
$$\therefore \angle ADE = 60^\circ, AD = AE = DE,$$

$$\therefore \angle BDE = \angle ADB + \angle ADE = 90^\circ.$$

$$\because BD = 1, BE = 2,$$

$$\therefore AD = DE = \sqrt{BE^2 - BD^2} = \sqrt{2^2 - 1^2} = \sqrt{3},$$

$\therefore AD$ 的长是 $\sqrt{3}$.



在 $\triangle FBE$ 和 $\triangle QBE$ 中,

$$\because BF=BQ, \angle FBE=\angle QBE, BE=BE,$$

$$\therefore \triangle FBE \cong \triangle QBE (\text{SAS}),$$

$$\therefore EF=EQ,$$

$$\therefore AE=EQ+AQ=EF+CF,$$

$$\text{即 } EF=AE-CF.$$